

Augmentation Intensity vs. Generalization in YOLOv8n for Iranian Cheque Detection Under Severe Data Scarcity

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Abstract—Data augmentation is a standard tool for improving the generalization of object detectors, but most augmentation recipes (mosaic, copy-paste, large-angle rotation, perspective warping) were tuned for natural-scene datasets with thousands of images per class. It is unclear whether the same augmentation intensity is appropriate for structured, rigid-layout documents such as bank cheques, particularly when the training set is extremely small. We present a controlled ablation study isolating augmentation aggressiveness as the only independent variable across four YOLOv8n training configurations – a minimal-augmentation baseline, a conservative document-safe configuration, an aggressive stress-test configuration, and a proposed balanced configuration – trained on a dataset of only 177 cheque images (141 train / 36 validation). All other hyperparameters (architecture, epochs, image resolution, batch size, and random seed) are held fixed. We find that the proposed balanced configuration achieves the best mAP@50 (0.9950), mAP@50-95 (0.6087), precision (0.9977), and recall (1.0000) of all four variants, simultaneously dominating the precision-recall trade-off. The heavy-augmentation variant exhibits the smallest train/validation generalization gap despite the worst raw mAP@50-95, suggesting augmentation intensity trades off optimization difficulty against overfitting resistance even at this small scale. These results support the use of moderate, layout-aware augmentation – rather than either extreme – for low-data structured-document detection tasks.

Index Terms—object detection, YOLOv8, data augmentation, ablation study, document detection, small-data learning, cheque recognition

I. Introduction

Automated detection and localization of financial documents, such as bank cheques, is a prerequisite step for downstream OCR and information-extraction pipelines in fintech and banking automation systems. Unlike natural-scene object detection, cheque detection operates on a single rigid-layout class: cheques have a near-constant aspect ratio, fixed printed fields, and limited legitimate pose variation (mild rotation and perspective from hand-held capture). At the same time, cheque image datasets are difficult to assemble at scale due to the sensitivity of financial documents, which means practitioners are

routinely forced to train detectors on datasets of only a few hundred images.

Modern detector training pipelines, including the Ultralytics YOLOv8 family used in this work, ship with augmentation recipes (mosaic tiling, copy-paste synthesis, mixup, large rotation/shear/perspective transforms) that were primarily validated on large natural-image benchmarks such as COCO. It is not obvious that the same augmentation intensity transfers to a single-class, structured-document setting with two orders of magnitude less data. Excessive geometric distortion may destroy the very layout cues that make a cheque recognizable, while too little augmentation may cause the detector to memorize the training set outright.

This paper isolates augmentation intensity as the sole independent variable in a four-way ablation study, training YOLOv8n on a fixed 177-image cheque dataset under four progressively different augmentation regimes. Our contributions are:

- A controlled ablation design that holds architecture, schedule length, image resolution, batch size, and seed fixed across all variants, isolating augmentation as the only manipulated factor;
- A proposed “balanced” augmentation configuration, tuned specifically for small, structured-document datasets, that empirically dominates both a minimal-augmentation baseline and a maximal-augmentation stress test on every evaluated metric;
- An analysis of the train/validation generalization gap across variants, showing that augmentation intensity and raw accuracy do not move together monotonically.

II. Related Work

YOLO-family detectors. The YOLO family of single-stage detectors [1] has evolved through successive versions to incorporate stronger backbones, anchor-free heads, and built-in augmentation pipelines. YOLOv8, used in this study, integrates mosaic augmentation, classification-style

HSV color jitter, and optional copy-paste/mixup strategies directly into its training loop.

Augmentation for small datasets. Mosaic augmentation, introduced in YOLOv4 [2], tiles four training images into one composite to expose the model to more object-scale and context variation per batch. Copy-paste augmentation [3] synthesizes additional positive instances by pasting segmented objects onto new backgrounds, and mixup [4] blends pairs of images and labels to regularize the decision boundary. All three techniques were developed and validated on large natural-image corpora; their behavior under severe data scarcity (under 200 images) and on rigid, single-class structured documents is comparatively unexplored.

Document image analysis. Document detection and layout analysis tasks typically deal with comparatively clean, structured inputs relative to natural scenes, motivating augmentation strategies that preserve layout integrity (e.g., avoiding aggressive shear or perspective warps that a real scanning or photography pipeline would not produce). This work positions itself at the intersection of these two lines of work: applying natural-image-derived augmentation primitives to a structured-document, low-data regime, and empirically measuring where the intensity dial should sit.

III. Dataset and Experimental Setup

A. Dataset

The dataset consists of 177 annotated cheque images, split 80/20 into 141 training and 36 validation images using a fixed random seed (Table I). A single object class, cheque, is annotated per image.

TABLE I
Dataset Composition

Split	Image Count
Training	141
Validation	36
Total	177

B. Common Training Configuration

All four variants share the configuration in Table II, training from pretrained YOLOv8n weights with identical architecture, schedule length, resolution, batch size, and seed. Only augmentation-related hyperparameters and their interacting optimization settings (learning rate, weight decay, patience, warmup, LR schedule) differ between variants, as detailed in Section IV.

IV. Methodology

Four configurations were designed to span a spectrum of augmentation intensity, from near-absent to deliberately excessive, with a fifth design point (the proposed balanced configuration) targeting the hypothesized sweet spot for small structured-document datasets.

TABLE II
Common Training Configuration (All Variants)

Parameter	Value
Model	YOLOv8n
Epochs (maximum)	120
Image size	640 × 640
Batch size	16
Random seed	42

A. Variant A – Baseline

Reproduces a minimal-augmentation configuration: mosaic probability suppressed to 0.3, no rotation, shear, or perspective warp, only mild scale/translate and HSV jitter. This establishes the lower bound of augmentation intensity tested.

B. Variant B – Light Augmentation

A conservative, document-safe configuration intended to respect the rigid layout of a cheque: low mosaic probability (0.1), no shear or perspective, small scale/translate, and a reduced learning rate (0.005) intended to guard against early overfitting on the small dataset.

C. Variant C – Heavy Augmentation

A deliberately aggressive stress test: full mosaic probability (1.0), strong HSV shifts, rotation up to 15°, shear, perspective warp, copy-paste, and mixup. This variant establishes the upper bound of augmentation-induced distortion and is expected to challenge the model’s ability to learn the comparatively rigid cheque layout.

D. Variant D – Balanced (Proposed)

The proposed configuration moderates mosaic probability to 0.5, applies moderate HSV jitter to absorb scan/lighting variation, introduces light copy-paste (0.1) to synthesize additional positive instances cheaply, and adds mild rotation (5°) without shear or perspective distortion. It additionally adopts cosine learning-rate decay and a longer warmup (5 epochs) to stabilize early gradient flow on the small dataset, with patience extended to 35 epochs to tolerate a longer plateau before early stopping.

Table III summarizes the augmentation-related hyperparameters for all four variants.

E. Evaluation Metrics

We report standard object detection metrics computed at each variant’s best checkpoint (selected by mAP@50 on the validation split): mean Average Precision at an IoU threshold of 0.50 (mAP@50), mean Average Precision averaged over IoU thresholds 0.50–0.95 in steps of 0.05 (mAP@50-95), precision, and recall. We additionally analyze per-epoch training and validation loss curves (box regression, classification, and distribution focal loss components) to characterize convergence behavior and overfitting.

TABLE III
Augmentation and Optimization Hyperparameters by Variant

Parameter	A: Baseline	B: Aug-Light	C: Aug-Heavy	D: Aug-Balanced
Initial LR (lr0)	0.008	0.005	0.010	0.006
Patience (epochs)	25	20	30	35
Weight decay	0.0005	0.0005	0.0010	0.0005
Mosaic probability	0.3	0.1	1.0	0.5
Close-mosaic (last N epochs)	10	5	15	10
Horizontal flip prob.	0.5	0.3	0.5	0.5
Scale jitter	0.5	0.2	0.9	0.4
Translate	0.1	0.05	0.3	0.15
HSV-Hue	0.015	0.01	0.05	0.02
HSV-Saturation	0.2	0.1	0.7	0.4
HSV-Value	0.2	0.1	0.5	0.3
Rotation (degrees)	0.0	0.0	15.0	5.0
Shear	0.0	0.0	5.0	0.0
Perspective	0.0	0.0	0.001	0.0
Copy-paste probability	0.0	0.0	0.2	0.1
Mixup probability	0.0	0.0	0.1	0.0
Cosine LR schedule	No	No	No	Yes
Warmup epochs	3	3	3	5

V. Results

A. Best-Epoch Performance

Table IV reports the best-epoch metrics achieved by each variant. The proposed balanced configuration (D) achieves the highest score on every metric: mAP@50 of 0.9950 (tied with the baseline), mAP@50-95 of 0.6087 (the best of all variants), precision of 0.9977, and recall of 1.0000. The heavy-augmentation variant (C) attains a competitive mAP@50 (0.9945) but the lowest mAP@50-95 (0.4808) of all four variants, indicating that while it can localize cheques coarsely, its bounding-box precision under stricter IoU thresholds degrades under aggressive geometric distortion. The light-augmentation variant (B) underperforms across all four metrics and converges earliest (best epoch 23 of a 120-epoch budget), consistent with the conservative learning rate and weak augmentation signal allowing early stopping to trigger before the model fully exploited the training set.

TABLE IV
Best-Epoch Metrics by Variant

Variant	mAP@50	mAP@50-95	Prec.	Recall
A: Baseline	0.9950	0.6077	0.9820	1.0000
B: Aug-Light	0.9791	0.5827	0.9136	0.9444
C: Aug-Heavy	0.9945	0.4808	0.9866	0.9722
D: Aug-Balanced	0.9950	0.6087	0.9977	1.0000

Figure 1 compares all four metrics side by side. Figure 2 plots best-epoch precision against recall for each variant; the balanced configuration sits strictly above and to the right of the light- and heavy-augmentation variants, dominating them on both axes simultaneously, and edges out the baseline on precision while matching it on recall.

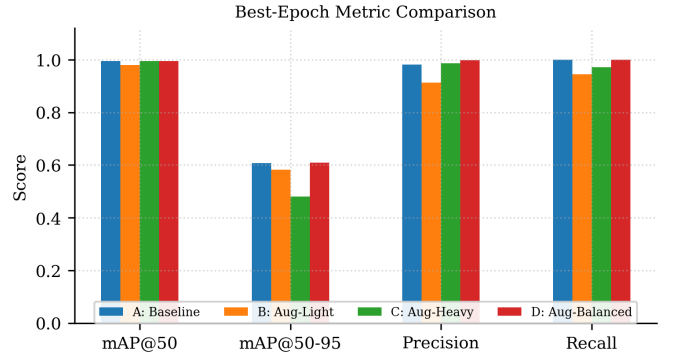


Fig. 1. Best-epoch metric comparison across all four augmentation variants.

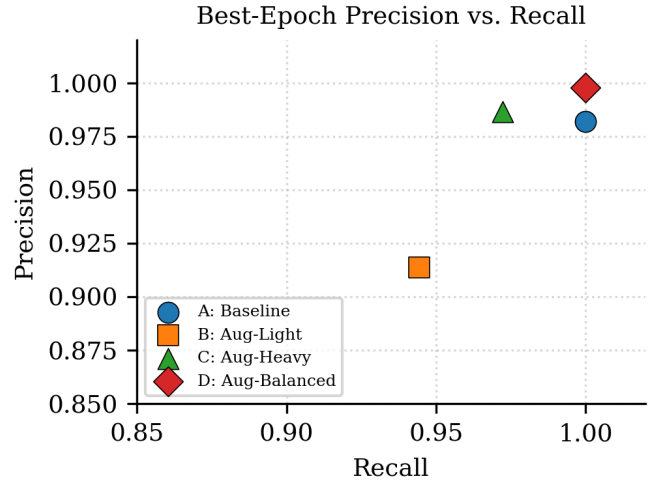


Fig. 2. Best-epoch precision vs. recall for each variant. The balanced configuration (D) is Pareto-dominant.

B. Convergence Behavior

Figures 3 and 4 show mAP@50 and mAP@50-95 over training, respectively. Note that variants stopped at different epochs under the early-stopping criterion: B at epoch 23, C at epoch 65, A at epoch 95, and D completed the full 120-epoch budget without triggering early stopping. The heavy-augmentation variant (C) converges visibly more slowly and noisily than the others on mAP@50-95, consistent with the harder learning signal introduced by aggressive geometric and photometric distortion, and never closes the gap to the baseline or balanced configuration within its training window.

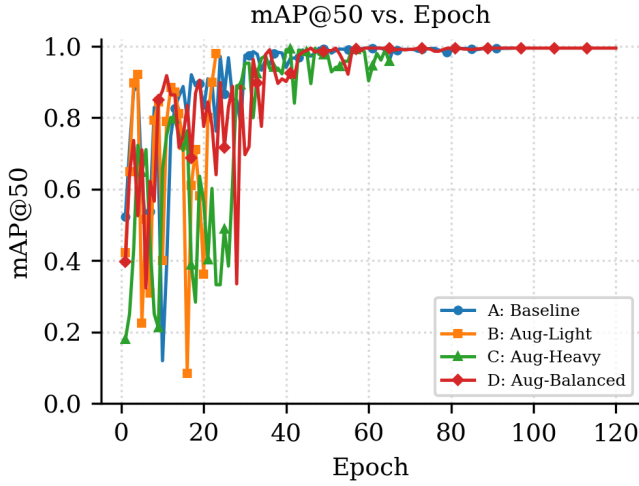


Fig. 3. mAP@50 vs. epoch for all four variants.

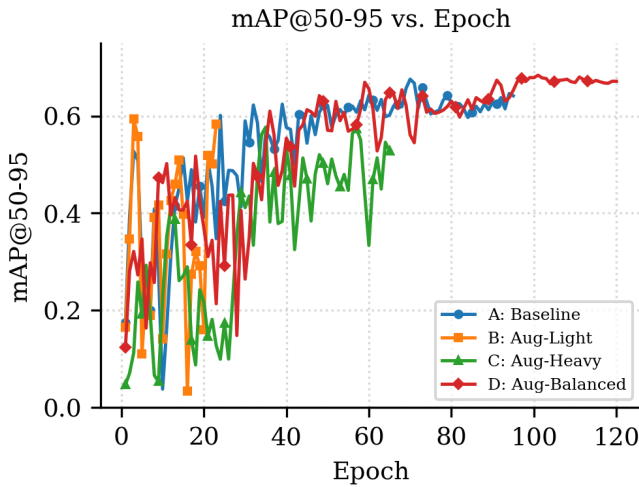


Fig. 4. mAP@50-95 vs. epoch for all four variants. The heavy-augmentation variant (C) plateaus markedly lower than the others.

C. Training Loss and Generalization Gap

Figure 5 shows training loss curves (box, classification, and distribution focal loss) for all variants. The heavy-

augmentation variant maintains a visibly higher training loss throughout, reflecting the increased difficulty of fitting heavily distorted inputs, while the baseline, light, and balanced variants track closely together with the lowest losses.

Figure 6 shows the generalization gap, computed as validation box loss minus training box loss, over the course of training. Counter-intuitively, the heavy-augmentation variant (C) exhibits the smallest generalization gap of all variants for most of training, despite its worse mAP@50-95. The baseline and balanced variants, by contrast, show a steadily widening gap as training progresses, indicating that they fit the small training set increasingly tightly even as held-out accuracy continues to improve up to their respective best epochs.

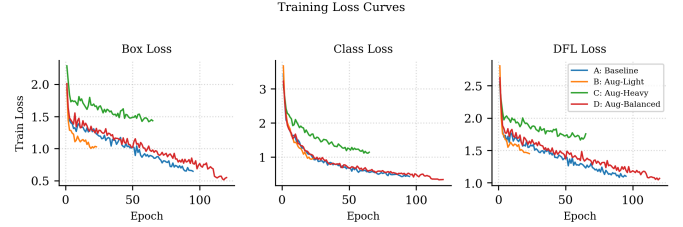


Fig. 5. Training loss curves (box, classification, DFL) for all four variants.

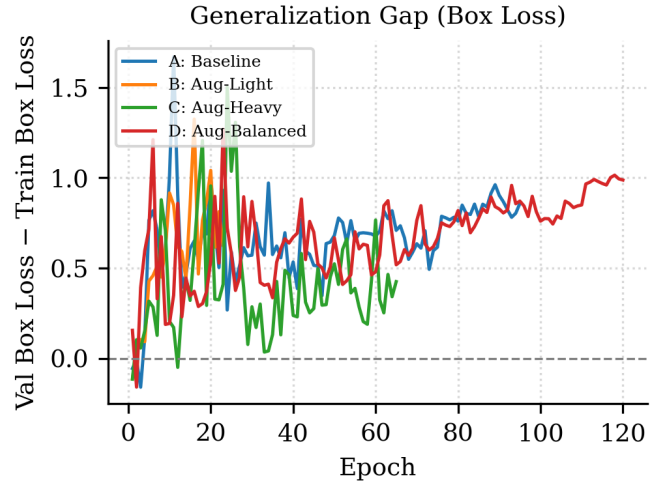


Fig. 6. Generalization gap (validation – training box loss) vs. epoch. Heavy augmentation (C) shows the smallest gap despite the lowest mAP@50-95.

VI. Discussion

The results support a non-monotonic relationship between augmentation intensity and detector quality on this small, structured-document dataset. Minimal augmentation (A) and the proposed balanced configuration (D) achieve the best raw accuracy, while heavy augmentation (C) sacrifices mAP@50-95 substantially despite matching

the others on mAP@50 – suggesting that aggressive distortion still allows the model to find the cheque (coarse localization, reflected in mAP@50 and recall) but degrades the precision of its bounding boxes (reflected in the mAP@50-95 drop and the lower IoU overlap implied by it).

The generalization-gap analysis (Figure 6) complicates a simple “more augmentation always overfits less, but converges worse” narrative only partially: heavy augmentation does show the smallest train/validation gap, consistent with its role as a regularizer, but this regularization comes at a direct cost to absolute accuracy at this dataset size. With only 141 training images, the model may simply not have enough signal to simultaneously absorb heavy geometric distortion and learn precise box regression within a 120-epoch budget.

The light-augmentation variant (B) underperforms not because its augmentation was too aggressive, but plausibly because the combination of weak augmentation signal and conservative learning rate caused premature convergence (best epoch 23) before the model had fully exploited the dataset – a reminder that augmentation intensity and optimization hyperparameters interact, and cannot be tuned in isolation.

Taken together, these results favor the balanced configuration’s design philosophy for this task: moderate mosaic probability, layout-preserving (shear- and perspective-free) geometric augmentation, light copy-paste synthesis, and a longer-patience cosine schedule. This configuration matches or exceeds the baseline on every metric while remaining far more robust than the heavy-augmentation extreme.

VII. Limitations

This study has several limitations that bound the generalizability of its conclusions. First, the dataset is small (177 images) and drawn from a single source distribution; results may not transfer to cheques from different countries, layouts, or scanning conditions. Second, only the smallest YOLOv8 variant (YOLOv8n) was evaluated; larger backbones (YOLOv8s/m) may respond differently to the same augmentation settings. Third, the four configurations differ in more than augmentation alone – learning rate, weight decay, patience, and LR schedule were co-tuned with each augmentation regime rather than held fixed – so the comparison reflects augmentation-centric configurations rather than a strict single-factor ablation of augmentation parameters alone. Finally, early stopping caused variants to train for different numbers of epochs, which should be considered when comparing convergence speed rather than final accuracy.

VIII. Conclusion and Future Work

We presented a controlled ablation study of augmentation intensity for YOLOv8n cheque detection under severe data scarcity (177 images). Across four configurations

spanning minimal to aggressive augmentation, a proposed balanced configuration – moderate mosaic probability, layout-preserving geometric transforms, light copy-paste synthesis, and a longer-patience cosine learning-rate schedule – achieved the best results on every evaluated metric (mAP@50 = 0.9950, mAP@50-95 = 0.6087, precision = 0.9977, recall = 1.0000), Pareto-dominating both the minimal-augmentation baseline and the aggressive stress-test configuration on the precision-recall trade-off. Heavy augmentation reduced the train/validation generalization gap but at a substantial cost to bounding-box precision, while light augmentation underperformed due to premature convergence. These findings suggest that augmentation recipes tuned for large natural-image datasets should not be applied at full intensity to small, structured-document detection tasks without adjustment.

Future work includes extending this ablation to larger YOLOv8 backbones and to additional structured-document classes beyond cheques, scaling the dataset to test whether the advantage of moderate augmentation persists as training data grows, and conducting a strict single-factor ablation that holds optimization hyperparameters fixed while varying only augmentation parameters, to more cleanly separate the two effects observed in this study. On-device deployment characteristics (TFLite float16 inference latency and accuracy degradation) of each variant are also left for future evaluation.

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